Supplementary Material: Four Biological Mechanisms Predict ITP Sex Differences from Male Data Alone

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S1 Full Parameter Tables

The ELM contains 168 parameters: 151 pathway rate constants and coupling coefficients (Tables S3–S10), 6 compound scale factors, 6 compound pathway ratios (Table S2), and 5 sex mechanism parameters (Table S11). The dual-action canagliflozin extension (Table S3) adds one parameter (the pathway split f), bringing the total to 169. These fall into five categories based on provenance.

Table S1: Parameter classification by provenance.

Category	Description	Count
Calibrated to ITP	Root-found compound scale factors (male ITP targets)	6
Literature	Published experimental measurements (PMIDs cited)	115
Set (plausible range)	Value placed within a biologically plausible range; constrained by mechanism knowledge but not derived from a specific measurement	18
Calibrated to control	Adjusted so the untreated mouse has realistic lifespan or damage trajectories; no intervention data used	6
Structural	Normalization constants, coupling topology, bounds, standard defaults	23

Only the 6 calibrated-to-ITP parameters are free. The remaining 162 are frozen priors: set before examining ITP lifespan outcomes and not adjusted during calibration. The 6 calibrated-to-control parameters were adjusted to produce a biologically realistic untreated mouse (for example,

correct lifespan, methylation trajectory, and damage accumulation rate) but were not informed by any ITP intervention data.

S1.1 Compound Pathway Ratios and Literature Basis

Each ITP compound perturbs one or more pathways. For multi-pathway compounds, the *ratio* of activity across pathways is fixed before calibration; only the overall scale factor α_c is fitted. Table S2 documents the literature basis for each ratio assignment.

Table S2: Compound pathway ratios and literature basis.

Compound	Pathway	Ratio	Basis
Rapamycin	mTORC1 inhib.	0.60	Rapamycin-FKBP12 directly inhibits mTORC1 at the FRB domain [Lamming et al., 2014]. AMPK activation is indirect: mTORC1 inhibition relieves S6K1-mediated negative feedback on IRS-1/PI3K, permitting LKB1→AMPK signaling. The 60:40 split is estimated from mechanism hierarchy (direct target vs. feedback effect); no publication quantifies the relative longevity contributions.
	AMPK	0.40	
Acarbose	AMPK	0.50	α-glucosidase inhibition reduces postprandial glucose, activating AMPK via increased AMP:ATP ratio. Gut microbiome remodeling (increased short-chain fatty acid production) reduces CD38-mediated NAD^+ consumption [Harrison et al., 2014]. The 50:50 split reflects two comparably important mechanisms; the literature establishes both pathways but does not quantify their relative longevity contributions.
	gut microbiome	0.50	

Compound	Pathway	Ratio	Basis
17α-Estradiol	AMPK	0.50	Non-feminizing estrogen that activates AMPK via ERα→LKB1 [McInnes et al., 2012] and reduces ROS via direct antioxidant properties of the phenolic A-ring. The 50:50 split reflects two well-documented mechanisms of comparable importance; no study isolates their relative contributions to lifespan.
	antioxidant	0.50	
Canagliflozin	AMPK	1.00	**Baseline model.** SGLT2 inhibition causes urinary glucose loss, activating AMPK via energy deficit. Single-pathway assignment ($\alpha_c = 0.16$). Underpredicts female efficacy by 4 pp (+5.0% vs. +9.0%).
Canagliflozin alternative (main text §3.1, Figure 3):			
Canagliflozin	antioxidant	0.88	**Dual-action model.** FGF21/ketogenesis via SGLT2-mediated glycosuria activates Nrf2 antioxidant defenses [Ferrannini et al., 2016, Osataphan et al., 2019, Packer, 2020]. Sweep-calibrated $f^* = 0.88$ ($\alpha_c = 0.60$). Female prediction +8.8% (0.2 pp error). One additional parameter identified by one female datum.
(dual-action)	AMPK	0.12	
Aspirin	antioxidant	0.50	COX-1/2 inhibition reduces prostaglandin-mediated ROS (antioxidant), suppresses NF-κB-driven inflammation (anti-inflammatory), and mildly activates AMPK via salicylate binding to the β1 subunit [Egan et al., 2004, Grosser et al., 2006]. The 50:30:20 split reflects the published mechanism hierarchy (COX inhibition is the dominant pharmacological effect, AMPK activation is secondary, and anti-inflammatory effects are tertiary) but the specific numbers are estimated, not measured.
	AMPK	0.30	

Compound	Pathway	Ratio	Basis
	anti-inflam.	0.20	
Glycine	antioxidant	0.90	Glycine is a glutathione precursor; oral supplementation increases glutathione synthesis and reduces oxidative stress markers. AMPK activation is minimal and indirect. The 90:10 split reflects glycine's primary role as an antioxidant substrate; the small AMPK component is estimated.
	AMPK	0.10	

Table S3: Canagliflozin model variants compared. The baseline model (AMPK-only) has 6 free parameters calibrated to male ITP data; all female predictions are out-of-sample. The dual-action model adds one parameter (f, the antioxidant fraction), identified by one female datum (canagliflozin $F = +9\%$). Male predictions are identical by construction.

	Baseline (AMPK-only)	**Dual-action**
Pathway ratios	AMPK 1.00	antioxidant 0.88 / AMPK 0.12
Scale factor α_c	0.16	0.60
Injection: AMPK	0.16	0.07
Injection: antioxidant	—	0.53
Free parameters	6	6 + 1 (f)
Female data used	none	canagliflozin only
Cana. male prediction	+14.0%	+14.0%
Cana. female prediction	+5.0%	+8.8%
Cana. female error	4.0 pp	0.2 pp
Mean female MAE (all 6)	1.3 pp	0.65 pp

S1.2 NAD$^+$ Homeostasis Parameters

Table S4: NAD$^+$ pathway parameters. $\Delta_{\max}$ defined in §S2.2.

Parameter	Value	Description	Source	$\Delta_{\max}$
$k_{\text{nampt,base}}$	0.55	Basal NAMPT activity	Yoshino et al. [2011]: rate-limiting enzyme	0.26
$k_{\text{nampt,ampk}}$	0.50	AMPK boost to NAMPT	Cantó et al. [2009]: AMPK activates NAMPT ∼1.5–2×	<0.01
$k_{\text{nampt,age}}$	0.431	Age-related NAMPT decline	Yoshino et al. [2011]: NAMPT declines with age	<0.1
$k_{\text{consumption}}$	0.40	Baseline NAD$^+$ turnover	Set within plausible range 0.25–0.55. Constrained by ∼50% NAD$^+$ decline over lifespan [Massudi et al., 2012]	0.17
$k_{\text{cd38,base}}$	0.35	Basal CD38 consumption	Chini et al. [2020]: major NADase	0.12
$k_{\text{cd38,age}}$	0.50	Age-related CD38 increase	Camacho-Pereira et al. [2016]: 2–3× with age	<0.1
$k_{\text{cd38,sasp}}$	0.60	SASP-induced CD38	Estimated from Covarrubias et al. [2020]: senescent cells upregulate CD38 via NF-κB. Plausible range 0.3–0.8	<0.1
k_{parp}	0.70	PARP consumption (damage-dep.)	Massudi et al. [2012]: PARP1 major consumer	0.12
k_{sirt}	0.10	Sirtuin consumption	SIRT1 uses NAD$^+$ as co-substrate [Imai et al., 2000]; minor consumer. Plausible range 0.05–0.20	<0.1

S1.3 AMPK/mTORC1 Signaling Parameters

Table S5: AMPK and mTORC1 pathway parameters. $\Delta_{\max}$: see §S2.2.

Parameter	Value	Description	Source	$\Delta_{\max}$
$K_{m,\text{ampk}}$	0.40	AMPK saturation constant	Set at midpoint of plausible range 0.2–0.8; AMPK saturation well-established (PMIDs 16469405, 23022416)	<0.01
$V_{\text{max,ampk}}$	1.0	Maximum AMPK activity	Structural (normalization)	<0.01
$k_{\text{mtorc1,auto}}$	0.70	mTORC1 inhib. → autophagy	Kim et al. [2011]: mTORC1 inhibition activates ULK1	<0.1
$k_{\text{mtorc1,sen}}$	0.35	mTORC1 inhib. → senescence	Demidenko et al. [2009]: rapamycin suppresses senescence	0.16
$\text{TSC2}_{\text{basal}}$	0.30	Basal TSC2 GAP activity	Huang and Manning [2008]	<0.1
$k_{\text{ampk,tsc2}}$	0.70	Max AMPK-induced TSC2	Inoki et al. [2003]: AMPK phosphorylates TSC2 at Thr1271	<0.01
$n_{\text{ampk,tsc2}}$	1.5	Hill coefficient (AMPK/GSK3)	Set at midpoint of plausible range 1.0–2.0; mild cooperativity (Inoki et al. [2003], PMID 16959574)	<0.01

S1.4 Sirtuin and Autophagy Parameters

Table S6: Sirtuin and autophagy parameters. $\Delta_{\max}$: see §S2.2.

Parameter	Value	Description	Source	$\Delta_{\max}$
$K_{m,\mathrm{sirt1}}$	0.4	NAD^+ affinity for SIRT1	SIRT1 K_m for NAD^+ ∼100–170 μM in vitro (PMID 10675329); normalized to 0.4. Plausible range 0.3–0.6	0.13
n_{sirt1}	2.0	SIRT1 Hill coefficient	Standard Hill coefficient for moderate sigmoidality ($n = 2$). Plausible range 1.0–3.0	<0.1
$k_{\mathrm{sirt1,foxo3}}$	0.8	SIRT1 → FOXO3 activation	Brunet et al. [2004]: SIRT1 deacetylates FOXO3	0.20
$V_{\mathrm{max,auto}}$	1.0	Maximum autophagy rate	Structural (normalization)	<0.1
$K_{m,\mathrm{auto}}$	0.35	Autophagy half-max	Set at midpoint of plausible range 0.2–0.5; autophagy has finite capacity (lysosomal processing, ATG proteins)	<0.1
$w_{\mathrm{ampk,auto}}$	0.45	AMPK contribution to autophagy	AMPK→ULK1 direct phosphorylation (Kim et al. [2011]); dominant autophagy input. Plausible range 0.3–0.6	<0.01
$w_{\mathrm{foxo3,auto}}$	0.35	FOXO3 contribution to autophagy	FOXO3 transcriptional autophagy program (Warr et al. [2013]). Plausible range 0.2–0.5	<0.1

S1.5 DNA Damage and Repair Parameters

Table S7: DNA damage and repair parameters. $\Delta_{\max}$: see §S2.2.

Parameter	Value	Description	Source	$\Delta_{\max}$
$k_{\text{damage,ros}}$	0.60	ROS-induced damage	Set to produce ∼3× damage accumulation over control lifespan (PMID 27026073). Plausible range 0.40–0.80	0.16
$k_{\text{damage,repl}}$	0.12	Replication errors	Constitutive replication damage [Lindahl, 1993]. Plausible range 0.05–0.25; minor contributor	<0.1
$k_{\text{damage,sasp}}$	0.25	SASP-induced damage	Coppé et al. [2008]: SASP paracrine damage	<0.01
k_{ber}	0.35	Base excision repair	Set so total repair (∼0.63) is exceeded by total damage (∼0.90). BER is PARP1-dependent (PMID 26924217). Plausible range 0.20–0.50	<0.1
$k_{\text{ber,nad}}$	0.7	BER NAD^+ dependence	Sub-linear NAD^+ dependence: PARP1 requires NAD^+ as substrate [Bai et al., 2011]. Plausible range 0.4–1.0	<0.1
$k_{\text{damage,sen}}$	0.15	Damage → senescence	Set within plausible range 0.05–0.30; secondary coupling via p53 (PMID 27026073)	<0.01
$k_{\text{antioxidant}}$	0.55	Antioxidant ROS reduction	Set to produce realistic control ROS dynamics (PMID 30048462). Plausible range 0.35–0.75	<0.01

S1.6 Methylation Parameters

Table S8: Methylation drift parameters. $\Delta_{\max}$: see §S2.2.

Parameter	Value	Description	Source	$\Delta_{\max}$
k_{ezh2}	1.85	Basal EZH2 activity	Set to produce realistic control methylation trajectory. Plausible range 1.5–2.2; dominant drift term	0.10
$k_{\text{ezh2,sirt1}}$	0.5	SIRT1 inhibition of EZH2	SIRT1 deacetylates EZH2, reducing H3K27me3 activity (Viré et al. [2006]). Plausible range 0.3–0.8	<0.01
$k_{\text{ezh2,ampk}}$	0.70	AMPK inhibition of EZH2	Viré et al. [2006]: AMPK→EZH2→DNMT	<0.01
k_{dnmt}	0.28	Basal DNMT activity	Constitutive DNMT1 maintenance methylation. Set within plausible range 0.15–0.40	<0.1
$k_{\text{sasp,meth}}$	0.42	SASP-induced methylation drift	Set at ∼40%, reflecting inflammation-associated CpG fraction in GrimAge (Lu et al. [2019]). Plausible range 0.2–0.6	<0.1
M_{young}	0.25	Youthful methylation level	Horvath [2013]: epigenetic clock baseline	<0.1

S1.7 Heteroplasmy Parameters

Table S9: Heteroplasmy dynamics parameters (Kowald–Kirkwood model). $\Delta_{\max}$: see §S2.2.

Parameter	Value	Description	Source	$\Delta_{\max}$
H_0	0.05	Initial heteroplasmy	Stewart and Chinnery [2015]: mtDNA mutation load at birth	<0.1
N_{eff}	100	Effective mtDNA population size	Shoubridge and Wai [2007]: bottleneck estimate	<0.1
α	0.20	Mutant transcription advantage	Wallace [2010]: deletion mutants replicate faster	**0.65**
s_0	0.001	Baseline selection	Negligible placeholder ensuring selection $\neq 0$ when FOXO3 $= 0$. Plausible range 0–0.01	<0.1
s_{foxo3}	0.10	FOXO3 selection boost	FOXO3→PINK1/Parkin mitophagy strength. Set within plausible range 0.05–0.20	**0.28**
s_{mito}	0.08	Mitophagy selection (60% clearance)	Calibrated to Pickrell et al. [2015] knockout data	<0.01
τ	12.0	Time scaling factor	Converts Kowald-Kirkwood time units to normalized lifespan. Set so H increases ∼3–4× (Wallace [2010]). Plausible range 8–18	**0.35**

S1.8 Senescence and SASP Parameters

Table S10: Senescence and SASP parameters. $\Delta_{\max}$: see §S2.2.

Parameter	Value	Description	Source	$\Delta_{\max}$
$k_{\text{sen,damage}}$	0.40	Damage-induced senescence	Campisi and d'Adda di Fagagna [2007]: DDR→senescence	<0.1
$k_{\text{sen,telomere}}$	0.12	Replicative senescence	Hayflick-limit arrivals (PMID 13905658). Set within plausible range 0.05–0.20; minor contributor	<0.1
$k_{\text{sen,clear}}$	0.02	Natural immune clearance	Baker et al. [2016]: low basal clearance	<0.1
$k_{\text{sen,auto}}$	0.70	Autophagy-mediated clearance	Autophagy facilitates immune surveillance of senescent cells (PMIDs 21979375, 24267888). Plausible range 0.40–1.00	<0.1
$k_{\text{ampk,sen}}$	0.45	AMPK protective effect	AMPK suppresses senescence entry via mTORC1 inhibition (PMID 18725386). Plausible range 0.30–0.60	<0.01
$k_{\text{sasp,prod}}$	0.45	SASP production rate	Coppé et al. [2008]: SASP dynamics	<0.1
$k_{\text{sasp,decay}}$	0.60	SASP decay rate	Aggregate SASP signal decay. Set within plausible range 0.30–1.00	<0.01

S1.9 BioAge and Normalization

Normalization constants are derived from the control simulation at $t = 1.0$ (see §S4). BioAge weights are swept separately in the weight sensitivity analysis (main text); normalization constants are structural.

Table S11: BioAge weights and normalization constants.

Parameter	Value	Description	Provenance	$\Delta_{\max}$
w_M	0.25	Methylation weight	Equal prior	—
w_D	0.25	DNA damage weight	Equal prior	—
w_H	0.25	Heteroplasmy weight	Equal prior	—
w_S	0.25	Senescence weight	Equal prior	—
$\hat{M}_0$	1.0925	Methylation at $t = 1.0$	Structural	—
$\hat{D}_0$	0.6653	Damage at $t = 1.0$	Structural	—
$\hat{H}_0$	0.1800	Heteroplasmy at $t = 1.0$	Structural	—
$\hat{S}_0$	0.2143	SenCells at $t = 1.0$	Structural	—

S1.10 Sex Mechanism Parameters

All sex-specific parameters are set from published measurements prior to examining female ITP data. They are not included in the OAT sweep because they apply only to one sex and are not re-calibrated.

Table S12: Sex-specific mechanism parameters.

Mechanism	Parameter	Value	Source
AMPK saturation	baseline b (F/M)	1.3	LKB1 expression [McInnes et al., 2012]
	curvature k	8	Midpoint of plausible range [3–20]; sensitivity sweep confirms mean female error 1.2–1.7 pp across entire range
	attenuation factor	0.29	Derived: $1/(1 + k(b - 1))$
CYP3A PK	exposure ratio (F/M)	3.48	Blood levels [Lamming et al., 2014]
Testosterone	female effect	0.0	Castration data [Garratt et al., 2018]
COX-2/PGI_2	female effect	0.0	HRT + NSAID data [Egan et al., 2004]

S2 Identifiability Analysis

S2.1 Calibration Jacobian

The calibration Jacobian $\mathbf{J}_{\text{cal}} \in \mathbb{R}^{12\times 6}$ has rows corresponding to the 12 compound-sex targets (6 male calibration + 6 female prediction) and columns corresponding to the 6 compound scale factors. Each entry $J_{ij} = \partial \Delta L^{(i)}/\partial \alpha^{(j)}$ was computed by finite differences (step $\epsilon = 10^{-3}$). The matrix has numerical rank 6, but aspirin's singular value is negligible (its small injection magnitudes produce near-zero cross-sensitivity to other scale factors), giving an effective rank of 5. Excluding aspirin, the remaining 5×5 block has condition number $\kappa \approx 1.0$: the calibration is well-posed, with 6 parameters determined by 6 independent male targets via root-finding.

S2.2 Sensitivity of Frozen Priors (One-at-a-Time Analysis)

Method. Each of the 141 perturbable frozen parameters was individually perturbed by ±10%. After each perturbation, the 6 compound scale factors were re-calibrated by root-finding so that male predictions still match ITP targets exactly. Because male predictions are held fixed by re-calibration, only perturbations that affect male and female mice *differently*, or that alter the combination's sub-additivity, produce nonzero shifts. All 6 female predictions and the rapa+acarbose combination prediction were then recomputed. The reported sensitivity is the maximum absolute shift across these 7 out-of-sample predictions.

Results. No parameter shifts any prediction by ≥1 pp. Of the 141 parameters swept, 15 (11%) produce shifts of 0.1–0.65 pp (moderate sensitivity) and 126 (89%) shift predictions by <0.1 pp (insensitive). Table S13 lists the 15 most sensitive frozen priors. The $\Delta_{\max}$ column in Tables S3–S10 reports this sensitivity for every parameter individually, so readers can verify the insensitivity claim directly.

Table S13: Top 15 most sensitive frozen priors.

Rank	Subsystem	Parameter	Value	Max shift (pp)
1	Heteroplasmy	transcription advantage α	0.20	0.65
2	Heteroplasmy	time scale τ	12.0	0.35
3	Heteroplasmy	FOXO3 selection s_{foxo3}	0.10	0.28
4	NAD^+	basal NAMPT k_{nampt}	0.55	0.26
5	mTORC1	damage reduction $k_{\text{mtorc1,dmg}}$	0.25	0.20
6	Sirtuin	SIRT1→FOXO3 $k_{\text{sirt1,foxo3}}$	0.80	0.20
7	TSC2	TSC2→mTORC1 max $k_{\text{tsc2,mtorc1}}$	0.65	0.20
8	NAD^+	basal turnover $k_{\text{consumption}}$	0.40	0.17
9	mTORC1	senescence effect $k_{\text{mtorc1,sen}}$	0.35	0.16
10	DNA repair	ROS damage $k_{\text{damage,ros}}$	0.60	0.16
11	Sirtuin	NAD^+ affinity $K_{m,\text{sirt1}}$	0.40	0.13
12	TSC2	TSC2→mTORC1 K_m	0.50	0.12
13	NAD^+	basal CD38 $k_{\text{cd38,base}}$	0.35	0.12
14	NAD^+	PARP consumption k_{parp}	0.70	0.12
15	Methylation	basal EZH2 k_{ezh2}	1.85	0.10

Why heteroplasmy dominates. Three of the top five parameters belong to the heteroplasmy subsystem. This is not an accident. The Kowald–Kirkwood equation for mitochondrial DNA dynamics is exponential: mutant mtDNA copies have a replicative advantage α that compounds over the animal's lifetime, so small changes in α or the time scale τ produce large shifts in the heteroplasmy trajectory. Unlike DNA damage, which has three parallel repair pathways (base excision repair, nucleotide excision repair, homologous recombination) that buffer perturbations,

heteroplasmy has only one clearance mechanism (FOXO3-driven mitophagy), leaving it with less compensating capacity. The NAD^+ and mTORC1 parameters that rank next (positions 4–9) sit upstream in long signal chains: NAMPT sets NAD^+ levels, which affect SIRT1, which affects FOXO3, which controls both autophagy and heteroplasmy selection. These long dependency chains propagate small perturbations.

Why most parameters are insensitive. The root-finding step is the key: when a frozen parameter is perturbed, the 6 compound scale factors adjust to compensate, holding male predictions fixed by construction. Female predictions only shift if the perturbation changes the *sex difference*: the gap between how male and female mice respond to the same compound. Most pathway parameters affect both sexes equally, so even substantial control-lifespan shifts are absorbed by recalibration.

Implications. Even the most sensitive frozen prior (α, ± 0.65 pp) shifts predictions by less than half the mean female prediction error (1.3 pp). This means prediction errors are dominated by model structure (which biological mechanisms are included or excluded, and how they are coupled) rather than by uncertainty in any individual parameter value.

S2.3 Is the Model a Disguised One-Parameter Fit?

If the model were secretly rescaling a single aging rate, then adjusting that rate would shift all predictions by the same fraction, and the compound scale factors would simply compensate. To test for this, we increased the heteroplasmy transcription advantage α by 10%. This shifts all 12 ITP predictions, but the fractional shift ratios vary by compound (0.151 for canagliflozin to 0.207 for acarbose, a 37% spread). If the model were merely fitting a rescalable rate, these ratios would be identical. The spread arises from nonlinear Michaelis–Menten saturation in the AMPK pathway, confirming genuine mechanistic content beyond rate rescaling.

S2.4 Leave-One-Out Analysis

For each compound, we asked: how well can the other five predict its behavior? We projected each compound's Jacobian sensitivity row onto the space spanned by the remaining five. The fraction captured measures how much of that compound's prediction is already determined by the others. All compounds are 79–99% captured:

- Glycine is most vulnerable (79.2%): its antioxidant-heavy profile is not shared by other compounds.
- 17α-Estradiol is most implied (99.9%): its male effect uses shared pathways; its female zero is categorical.
- Estimated leave-one-out errors are bounded within $\sim$5% for all compounds.

S3 BioAge Weight Validation Against Published Clocks

To assess whether equal BioAge weights ($w = 0.25$ per subsystem) are consistent with published biological age clocks, we classified the component biomarkers of three clocks into ELM-equivalent categories and summed their published weights (Table S14).

Classification procedure. Each biomarker in GrimAge [Lu et al., 2019], PhenoAge [Levine et al., 2018], and DunedinPACE [Belsky et al., 2022] was assigned to the ELM subsystem it most directly reflects. For example: GrimAge's DNAm-estimated PAI-1 (plasminogen activator inhibitor-1) and TIMP-1 are products of the senescence-associated secretory phenotype, so they map to Senescence/SASP. DNAm-estimated smoking pack-years reflects cumulative oxidative and mutagenic damage, mapping to DNA Damage. GrimAge's age and sex terms were excluded (not biological subsystems). PhenoAge's albumin and creatinine reflect metabolic/mitochondrial function; its CRP and lymphocyte percentage reflect inflammation/senescence. DunedinPACE is a pace-of-aging composite; we used the CpG site annotations from Belsky et al. [2022] Table S2 to classify by nearest biological pathway.

Where a biomarker plausibly spans two categories (for example, glucose reflects both metabolic and DNA damage pathways), we split its weight equally. Biomarkers with no clear ELM mapping (for example, white blood cell count in PhenoAge) were assigned to an "Other/unmapped" category and excluded from the per-subsystem percentages. Ranges in the table reflect classification uncertainty (alternative plausible assignments shift shares by ±5%).

Table S14: ELM subsystem shares from three biological age clocks.

Subsystem	GrimAge	PhenoAge	DunedinPACE	ELM weight
Heteroplasmy / mitochondrial	10–20%	15–25%	10–20%	25%
DNA damage	15–25%	20–30%	15–25%	25%
Methylation / epigenetic drift	20–30%	5–15%	25–30%	25%
Senescence / SASP	15–25%	15–25%	10–20%	25%

The decomposition does not pin down a unique weight vector, but it excludes pathological ones. Setting $w_M = 0.50$, for example, exceeds the range implied by both GrimAge (20–30%) and PhenoAge (5–15%). All four equal weights (0.25) fall within every clock's implied range. This does not prove the weights are correct; it establishes that equal weights are consistent with available clock data, and that extreme alternatives are not.

S4 Normalization Procedure

Normalization constants $(\hat{H}_0, \hat{D}_0, \hat{M}_0, \hat{S}_0)$ are derived from the control (no-intervention) simulation evaluated at $t = 1.0$ (median lifespan). This ensures BioAge(1.0) = 1.0 by construction. The ODE dynamics are independent of these normalization constants; they serve only to set the

death-threshold scale. Changing BioAge weights therefore changes the relative contribution of each subsystem at death without introducing circularity, because the normalization constants are recomputed for each weight vector via `derive_normalization_constants()`.

S5 Integration and Numerical Details

ODEs were integrated with SciPy's `solve_ivp` using the RK45 method (rtol = 10^{-8}, atol = 10^{-10}). Compound scale factors were calibrated by SciPy `brentq` root-finding (xtol = 10^{-3}). Convergence was verified by halving the integration step size and confirming predictions shifted by < 0.01 pp.

Sensitivity analysis used one-at-a-time perturbation ($\pm 10\%$) with re-calibration of all 6 compound scale factors per perturbation (141 parameters, 19 minutes on a single core).

All code, calibrated parameter sets, and figure generation scripts are available at `https://github.com/silmonbiggs/extensible-longevity-model` (MIT license).

References

Dudley W. Lamming, Maria M. Mihaylova, Pekka Katajisto, et al. Depletion of Rictor, an essential protein component of mTORC2, decreases male lifespan. *Aging Cell*, 13(5):911–917, 2014. PMID 25059582. CYP3A sex difference: PMID 12727820.

David E. Harrison, Randy Strong, David B. Allison, et al. Acarbose, 17-α-estradiol, and nordihydroguaiaretic acid extend mouse lifespan preferentially in males. *Aging Cell*, 13(2):273–282, 2014. PMID 24245565.

Karen J. McInnes, Kerry A. Brown, Natalie I. Hunger, and Evan R. Simpson. Regulation of LKB1 expression by sex hormones in adipocytes. *International Journal of Obesity*, 36(7):982–985, 2012. PMID: 21876548.

Ele Ferrannini, Simona Baldi, Silvia Frascerra, Brenno Astiarraga, Tim Heise, Roberto Bizzotto, Andrea Mari, Thomas R Pieber, and Elza Muscelli. Shift to fatty substrate utilization in response to sodium-glucose cotransporter 2 inhibition in subjects without diabetes and patients with type 2 diabetes. *Diabetes*, 65(5):1190–1195, 2016. PMID: 26861783.

Soravis Osataphan, Chiara Macchi, Garima Singhal, Jeremy Chimene-Weiss, Vicencia Sales, Chisayo Kozuka, Jonathan M Dreyfuss, Hui Pan, Yanin Tangcharoenpaisan, Jordan Morningstar, Robert Gerszten, and Mary-Elizabeth Patti. SGLT2 inhibition reprograms systemic metabolism via FGF21-dependent and -independent mechanisms. *JCI Insight*, 4(5):e123130, 2019. PMID: 30843877.

Milton Packer. Role of ketogenic starvation sensors in mediating the renal protective effects of SGLT2 inhibitors in type 2 diabetes. *Journal of Diabetes and its Complications*, 34(9):107647, 2020. PMID: 32534886.

Kathleen M. Egan, John A. Lawson, Susanne Fries, et al. COX-2-derived prostacyclin confers atheroprotection on female mice. *Science*, 306(5703):1954–1957, 2004. PMID 15550624.

Tilo Grosser, Susanne Fries, and Garret A. FitzGerald. Biological basis for the cardiovascular consequences of COX-2 inhibition: therapeutic challenges and opportunities. *Journal of Clinical Investigation*, 116(1):4–15, 2006. PMID 16395396.

Jun Yoshino, Kathryn F. Mills, Myeong Jin Yoon, and Shin-ichiro Imai. Nicotinamide mononucleotide, a key NAD^+ intermediate, treats the pathophysiology of diet- and age-induced diabetes in mice. *Cell Metabolism*, 14(4):528–536, 2011. PMID 21982712.

Carles Cantó, Zachary Gerhart-Hines, Jerome N. Feige, et al. AMPK regulates energy expenditure by modulating NAD^+ metabolism and SIRT1 activity. *Nature*, 458(7241):1056–1060, 2009. PMID 19262508.

Hassina Massudi, Ross Grant, Nady Braidy, et al. Age-associated changes in oxidative stress and NAD^+ metabolism in human tissue. *PLOS ONE*, 7(7):e42357, 2012. PMID 22848760.

Claudia C. S. Chini, Thais R. Peclat, Gina M. Warner, et al. CD38 ecto-enzyme in immune cells is induced during aging and regulates NAD^+ and NMN levels. *Nature Metabolism*, 2(11):1284–1304, 2020. PMID 33199925.

Juliana Camacho-Pereira, Mariana G. Tarragó, Claudia C. S. Chini, et al. CD38 dictates age-related NAD decline and mitochondrial dysfunction through an SIRT3-dependent mechanism. *Cell Metabolism*, 23(6):1127–1139, 2016. PMID 27304511.

Anthony J. Covarrubias, Abhijit Kale, Rosalba Perber, et al. Senescent cells promote tissue NAD^+ decline during ageing via the activation of $CD38^+$ macrophages. *Nature Metabolism*, 2(11):1265–1283, 2020. PMID 33199925.

Shin-ichiro Imai, Christopher M. Armstrong, Matt Kaeberlein, and Leonard Guarente. Transcriptional silencing and longevity protein Sir2 is an NAD-dependent histone deacetylase. *Nature*, 403 (6771):795–800, 2000. PMID 10693811.

Joungmok Kim, Mondira Kundu, Deepa P. Bhatt, et al. AMPK and mTOR regulate autophagy through direct phosphorylation of Ulk1. *Nature Cell Biology*, 13(2):132–141, 2011. PMID 21258367.

Zoya N. Demidenko, Stella G. Zubova, Elena I. Bukreeva, Valery A. Pospelov, Tatiana V. Pospelova, and Mikhail V. Blagosklonny. Rapamycin decelerates cellular senescence. *Cell Cycle*, 8(12):1888–1895, 2009. PMID 19471117.

Jingxiang Huang and Brendan D. Manning. The TSC1–TSC2 complex: a molecular switchboard controlling cell growth. *Biochemical Journal*, 412(2):179–190, 2008. PMID 18466115.

Ken Inoki, Tianqing Zhu, and Kun-Liang Guan. TSC2 mediates cellular energy response to control cell growth and survival. *Cell*, 115(5):577–590, 2003. PMID 14651849.

Anne Brunet, Lorraine B. Sweeney, John F. Sturgill, et al. Stress-dependent regulation of FOXO transcription factors by the SIRT1 deacetylase. *Science*, 303(5666):2011–2015, 2004. PMID 14976264.

Matthew R. Warr, Mikhail Binnewies, Johanna Flach, et al. FOXO3A directs a protective autophagy program in haematopoietic stem cells. *Nature*, 494(7437):323–327, 2013. PMID 23389440.

Tomas Lindahl. Instability and decay of the primary structure of DNA. *Nature*, 362(6422):709–715, 1993. PMID 8469282.

Jean-Philippe Coppé, Christopher K. Patil, Francis Rodier, et al. Senescence-associated secretory phenotypes reveal cell-nonautonomous functions of oncogenic RAS and the p53 tumor suppressor. *PLOS Biology*, 6(12):2853–2868, 2008. PMID 19053174.

Péter Bai, Carles Cantó, Hugues Oudart, et al. PARP-1 inhibition increases mitochondrial metabolism through SIRT1 activation. *Cell Metabolism*, 13(4):461–468, 2011. PMID 21459330.

Emmanuelle Viré, Carmen Brenner, Rachel Deplus, et al. The Polycomb group protein EZH2 directly controls DNA methylation. *Nature*, 439(7078):871–874, 2006. PMID 16357870.

Ake T. Lu et al. DNA methylation GrimAge strongly predicts lifespan and healthspan. *Aging*, 11 (2):303–327, 2019. PMID 30669119.

Steve Horvath. DNA methylation age of human tissues and cell types. *Genome Biology*, 14:R115, 2013. PMID 24138928.

James B. Stewart and Patrick F. Chinnery. The dynamics of mitochondrial DNA heteroplasmy: implications for human health and disease. *Nature Reviews Genetics*, 16(9):530–542, 2015. PMID 26281784.

Eric A. Shoubridge and Timothy Wai. Mitochondrial DNA and the mammalian oocyte. *Current Topics in Developmental Biology*, 77:87–111, 2007. PMID 17222701.

Douglas C. Wallace. Mitochondrial DNA mutations in disease and aging. *Environmental and Molecular Mutagenesis*, 51(5):440–450, 2010. PMID 20544884.

Alicia M. Pickrell, Chun-Hung Huang, Scott R. Kennedy, et al. Endogenous Parkin preserves dopaminergic substantia nigral neurons following mitochondrial DNA mutagenic stress. *Neuron*, 87(2):371–381, 2015. PMID 26182419.

Judith Campisi and Fabrizio d'Adda di Fagagna. Cellular senescence: when bad things happen to good cells. *Nature Reviews Molecular Cell Biology*, 8(9):729–740, 2007. PMID 17667954.

Darren J. Baker, Bennett G. Childs, Matej Durik, et al. Naturally occurring p16(Ink4a)-positive cells shorten healthy lifespan. *Nature*, 530(7589):184–189, 2016. PMID 26840489.

Michael Garratt, Katy A. Lagerborg, Yu-Miin Tsai, Andrzej Galecki, Mohit Jain, and Richard A. Miller. Male lifespan extension with 17-α estradiol is linked to a sex-specific metabolomic response modulated by gonadal hormones in mice. *Aging Cell*, 17(4):e12786, 2018. PMID 29806096.

Morgan E. Levine et al. An epigenetic biomarker of aging for lifespan and healthspan. *Aging*, 10 (4):573–591, 2018. PMID 29676998.

Daniel W. Belsky et al. DunedinPACE, a DNA methylation biomarker of the pace of aging. *eLife*, 11:e73420, 2022. PMID 35029144.